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Resource planning in the emergency departments: A simulation-based metamodeling approach

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ABSTRACT

Patient's congestion and their long waiting times in Emergency Departments (EDs) are the most common problems in hospitals. This paper extends application domain of metamodels into decision-making in the EDs by developing a discrete event simulation (DES) model combined with suitable metamodels. This is used as a novel decision support system to improve the patients flow and relieve congestion by changing the number of ED resources (i.e., the number of receptionists, nurses, residents, and beds). This new tool could be used for decision-making in operational, tactical, and strategic levels. In the first step, we develop a simulation routine of the ED in order to evaluate the system performance measure (total average waiting times of patients) for each configuration of resources. In the next step, we use different metamodel techniques and choose one with the maximum efficiency through a cross validation technique to replace the computationally expensive DES model with an accurate and efficient metamodel. Then the proposed model is used to minimize the total average waiting times of patients subject to both budget and capacity constraints. We implement our proposed model in an emergency department in Iran. Experimental results with the current ED budget verify that after using the resource allocation obtained from the proposed model, the total waiting time of patients is reduced by about 48%. Furthermore, to evaluate the efficiency of the selected metamodel, we compare the respective results with those obtained through OptQuest in terms of both the accuracy and the time needed to perform the optimization process.

1. Introduction

In recent years, the pressure on the Emergency Departments (EDs), as the main point of entry into hospitals, increased due to remarkable growth in demand (for various reasons e.g., aging population and trend toward utilizing the ED for nonemergency care), low productivity, and the limited budget and resources which mostly led to overcrowding and long queues in EDs [1–3]. As it was reported in the United States [4], ED visits went up about 20% between 1995 and 2005 while 381 EDs have been closed because they were not economically justified and the insurance companies could not repay their liabilities.

This overcrowding generally creates many intractable troubles including excessive number of patients, ambulance diversions, long waiting times, leaving without treatment, and inflated staff workload which disturb the process of serving the

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current patients, endangers patients' lives, increase the rate of dissatisfaction, and decreases the respective performance measures in EDs [5,6].

To ensure availability of resources and provide safe and quality care, the Ministry of Health and Medical Education in Iran proposed an incentive plan for governmental EDs. The ministry also obligates EDs to deliver their services in a timely manner. So, each patient should receive the required services and be discharged or admitted to inpatient care units within 6 h after his/her arrival. To reach such a goal, resource (staff, beds, etc.) planning plays an important role.

Making decisions about ED's resources is a challenging activity that has significant impact on EDs performance [7]. Any incorrect decisions can have serious consequences on the quality of services. The decision-maker should analyze the system and allocate the hospital resources more efficiently. This analysis is not straightforward since EDs are stochastic environments and have time-dependent behavior (mainly variations in the patient's arrival rates and service rates). So monitoring the state of EDs and planning their resources in strategic, tactical and operational level can help ED managers to enhance system performance and reach to predefined standards.

Since EDs are quite complex systems and it is difficult to model thorough analytical methods, computer simulation widely used for modeling of these systems [8,9]. Simulation is adopted to imitate the current state of any process. Many of these applications are reviewed in Paul et al. [10]. Simulation enables decision makers to imitate the behavior of the system and provides a test bed to assess changes in operations and managerial policies and examine different alternatives.

However, the simulation is not an optimization technique and to find the optimum value of decision variables, simulation-based optimization methods should be used [11]. In recent years, the iterative methods, especially a combination of simulation models with heuristics are used for planning and optimization in healthcare systems. Baesler and Sepúlveda [12] present a multi-objective simulation optimization model for improve patient flow in a cancer treatment center. They used genetic algorithm and goal programming to optimize the design of system. Ahmed and Alkhamis [13] integrated simulation with optimization to design a decision support tool for the operation of an ED at governmental hospital in Kuwait. The simulation-based-optimization model presented in this paper provides optimal resource allocation that would allow a 28% increase in patient throughput and an average of 40% reduction in patients' waiting time with the same resources. Cabrera et al. [14] presented an Agent-Based modeling and simulation to design a decision support system (DSS) for the operation of healthcare ED. Optimization is performed by exhaustive search to find the configuration of ED staff configuration of doctors, triage nurses, and admission personnel that simultaneously minimizes patients waiting time and maximizes patients throughput. Lin et al. [15] used a design point's relative efficiency score from data envelopment analysis (DEA) as its fitness value in the selection operation of genetic algorithm (GA) to determine optimal resource configuration in surgical services. Al-Refaie et al. [16] applied the integration of simulation and DEA in the emergency department to find out the best distribution of nurses in order to increase utilization and decrease patient waiting time.

Although these methods have been successfully used in different problems, they usually encounter two main issues: (1) developing a suitable heuristics and their tuning with a simulator are generally troublesome (2) exploring the solution space and optimizing a simulator takes a lot of time especially when a real-life problem is going to be tackled. To cope with these problems and to create a proper decision support system (DSS) for planning the scarce resources of the emergency departments we construct an approximation model of simulation model in which the relationship between the input and output of the simulation model is appropriately captured by certain sampling points. Kleijnen [17] called such approximation models "metamodels". As the mathematical curvature of the metamodels is usually smooth and there is no need to evaluate the underlying function using computationally expensive simulation which takes a lot of time, the optimization process could be implemented much faster than the other simulation based optimization models. Therefore, simulation-based-metamodels could potentially provide a viable platform for decision making especially in operational and tactical levels. To get more information about metamodels, one can refer to Kleijnen and Sargent [18] and Barton and Meckesheimer [19].

In this paper, we propose a fast and flexible DSS using simulation-based metamodels to obtain the optimal configuration of ED resources which minimize the total patient waiting times with consideration of budget and capacity constraints. The proposed decision support system can be used for creating guidelines in operational, tactical and strategic management practices. Simulation-based-metamodeling optimization process as the core part of this decision support system has been recently used in some practical cases especially where the time and the respective cost for reaching a feasible or optimal solution is really important [20,21]; however, it is a new topic in health care resources planning problems in which the time for taking tactical and operational decisions is really important. In other words, this is a pioneering research study that can be applied and extended for other complex health care problems in which all related departments are integrated to each other. We expanded the application area of these techniques into decision-making in emergency departments where the stochastic nature of governor parameters causes queues to build up frequently and decisions need to be made quickly even in operational level (a few hours later) to relieve the congestion and resultant problems. Based on our knowledge, there are few papers related to operational decision making in EDs and creating a real-time command-and-control solution is yet the challenge [22,23].

We believe that by continuous monitoring and control of the ED's state, considering stochastic parameters, proper prediction of the state of EDs finding the optimum design, and balancing the patient flow over the system (before congestion), the waiting times can be remarkably reduced and the resource utilization can be dramatically enhanced.

The remainder of the paper is organized as follows: in Section 2, we provide a detailed description of the emergency department, basic elements of the simulation model, and the problem formulation. The proposed metamodel-based optimization is presented in Section 3. Experimental results and comparing those with other well-known methods are given in Section 4. Section 5 is devoted to presenting a conceptual framework for operational decision-making in EDs. Finally, Section 5 contains conclusions and few suggestions for future research studies.

2. Modarres emergency department

2.1. System description

Modarres Hospital is located in the north of Tehran. The emergency department of Modarres Hospital is open 24 h a day and receives an average of 52,000 patients a year (average of 145 patients daily). The ED center consists of two main sections, "Chest Pain Unit (CPU)" for patients with heart problems and "General" for other patients. This separation was performed to provide better service delivery for patients with heart problems. About 40% of patients that enter this ED have heart problems and it is critical for them to receive their services as soon as possible. Since resource planning to reduce total average waiting times of patients is more critical for CPU, we considered this section in our simulation. The key resources in the ED are as follow:

1. Triage nurses (x_1)
2. Receptionists (x_2)
3. Nurses (x_3)
4. Heart residents (x_4)
5. Beds (x_5)

Currently ED has one triage nurse, two receptionists, three nurses, one heart resident, and seven monitoring beds. Patients arriving at the ED follow a process depicted in Fig. 1. The process begins when a patient arrives through the ED entrance door and ends when a patient is discharged from the ED or admitted into the hospital inpatients units. Based on the Emergency Severity Index (ESI) standard [24], patients are categorized from 1 (most urgent) to 5 (least urgent) levels by triage nurses. The triage system identifies: (1) the condition of patients (2) pathway each patient type goes through and (3) the corresponding resources. ESI 1 is related to the patients with urgent conditions who usually come to the ED with an ambulance. These patients promptly enter CPR (Cardio Pulmonary Resuscitation) where they can be resuscitated. If CPR action is successful, the patient is transferred to CPU for further treatment. The other four types of patients go through the triage nurse who assesses their acuity. Then, ESI 2 patients skip the reception step without waiting and directly go to CPU. ESI 3, 4, and 5 should go through the receptionist who collects the patient's personal information and locates their files. After the reception, ESI 3 and ESI 4 patients go to the CPU to receive their services and ESI 5 patients are discharged from ED after being visited by the general physician in ED.

Once a patient enters the CPU, a bed is assigned and ECG (Electrocardiography) which measures the electrical conduction system of the heart is taken by a nurse. Following this, the patients wait for availability of heart residents who should decide whether the treatment is enough or not. If the treatment is enough, patients should leave the ED, or should be admitted to an inpatient care unit. If not, patients receive further treatment (such as lab tests, medical advice, monitoring and another ECG after a few hours). It is worth noting that there might be a queue for receiving different services except for CPR. The main reason is that when the ESI 1 patient enters CPR, even seconds are vital for saving his/her life; therefore, all predetermined staff should temporarily interrupt their tasks and give the needed services to this patient.

2.2. Simulation model

The simulation model for the described system was constructed using the simulation package Arena 14.0 [25]. The key inputs required for simulation model are (1) the frequency of each pathway (2) arrival rates of patients and (3) duration of services. To estimate these inputs properly, we collected available patient's records from archived files from 20 February 2014 to 20 March 2014. The total number of data collected was about 4300 where 2% of them are ESI 1, 20% are ESI 2, 35% are ESI 3 and ESI 4, and 43% are ESI 5. Around 80% of data was used as input data which is fed into the simulation model and another 20% that is used for model validation. Based on these data, the arrival process was fitted with a non-homogenous Poisson process with rate $\lambda(t)$. $\lambda(t)$ denotes the estimated arrival rate of patients to the ED in time t. The arrival time interval set starts from 00.00–1:00 and ends at 23:00–24:00 at an hourly basis, that is, there are a total of 24 intervals (Fig. 2).

Statistical distributions for length of services were estimated by the arena input analyzer (Table 1). The distributions with the highest goodness of fit are selected.

We ran the simulation model for 60 days, 2 days warm-up period, and 100 replicated times. The warm-up period is set for the simulation run to eliminate any bias at the early stages of the process. Since the system is always initialized between replications so the confidence intervals of the simulation outputs can be calculated at 95% confidence level ($\alpha = 0.05$). We use these confidence intervals for comparing the simulation outputs by the actual ones. In order to approximate the number of replications, the average half-width of each confidence interval for all performance metrics is calculated by trial-and-error approach until that it is no more than 5% of the average mean and the length of running time of the simulation is tractable.

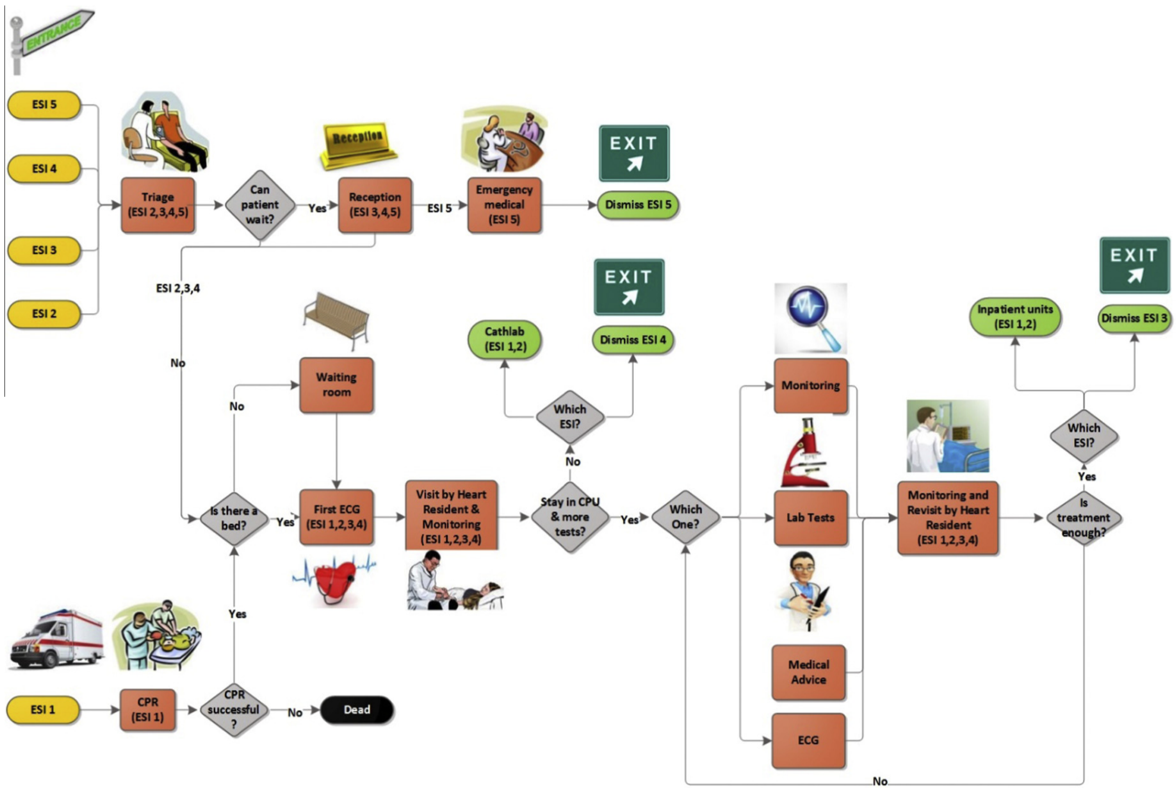

Fig. 1. Process chart.

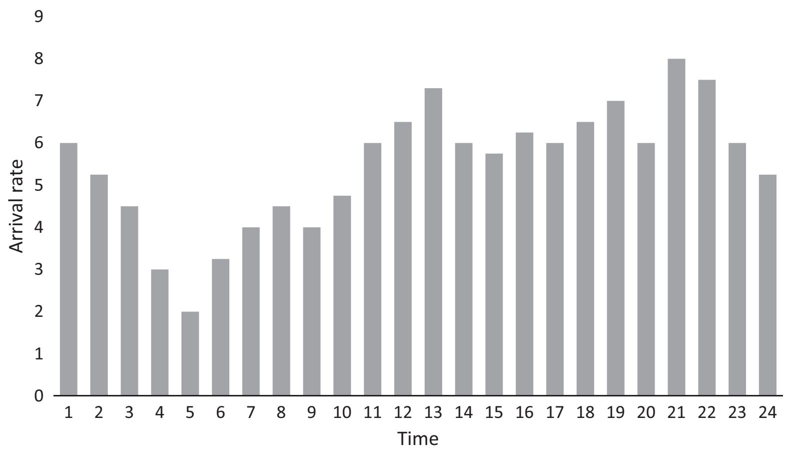

Fig. 2. Plot of the estimated arrival rate of patients.

2.3. Verification and validation

Verification is the task of ensuring that the model behaves as intended. The simulation model has been verified by the hospital's managers. They confirmed that the outputs of this model are reliable sources used for observing and monitoring the real situation of the Modarres ED. This confirmation has been accomplished by comparing the animation module created by Arena with what ED's managers experience in reality.

Moreover, the final results should be validated using a different set of collected data which have not been used in the simulation process. The main reason of validation is to check whether or not the simulation model is an accurate representation of the real system [26].

Table 1
Service time distributions at each stage.

Stage	Distribution (min)	Kolmogorov–Smirnov (*p*-values)
Triage	Uniform (2, 5)	0.091
Reception	Uniform (3, 5)	0.135
Monitoring	Uniform (30, 60)	0.105
CPR	Triangular (30, 45, 60)	0.078
ECG	Triangular (5, 10, 15)	0.214
Test process	Triangular (10, 15, 20)	0.192
Visit process	Triangular (10, 15, 20)	0.207
Lab tests	Triangular (45, 90, 180)	0.159
Medical advice	Triangular (15, 45, 90)	0.112

Since the total time spent in the system was recorded more precisely (in the hospital information system by receptionists), we used this data for validation. Table 2 shows confidence intervals of the simulation outputs at 95% ($\alpha = 0.05$) confidence level and the actual values obtained from the collected data. The comparison verifies that for each patient type, there were no significant differences between the results obtained using the simulation and those which occurred in the real system.

This could be an interest point that the total number of patients left the ED in one month is about 4500 patients based on simulation model, while the actual number of patients according to the hospital statistics is about 4350 patients. This might be another reason showing the validity of our simulation model.

2.4. Resource planning problem

As it is mentioned above, the ED managers would like to find the configuration of the key resources such that it minimizes the patient waiting times subject to budget and capacity constraints. The optimization problem can be mathematically expressed as follows:

$$\text{Min } Z = f(x_1, x_2, \ldots, x_n), \tag{1}$$

s.t.

$$\sum_{i=1}^{5} c_i x_i \leqslant B, \tag{2}$$

$$l_i \leqslant x_i \leqslant u_i \quad \text{for } i = 1, \ldots, 5, \tag{3}$$

$$x_i \text{ integer} \quad \text{for } i = 1, \ldots, 5. \tag{4}$$

This problem is a discrete optimization problem where Z represents the total average waiting time of patients which is a function of the decision variables $(x_1, x_2, \ldots, x_5)$. This function does not have analytical form and should be therefore evaluated by the simulation model described in the previous section. c_i is the monthly cost of each resource, B is the monthly budget value and l_i and u_i are the minimum and maximum capacity levels, respectively. All values of these parameters which were determined by managerial considerations are illustrated in Table 3.

3. Metamodel based optimization

3.1. Problem definition

The mathematical form of a stochastic simulation-based optimization problem can be expressed as follows:

$$\min_{\theta \in \Theta} f(\theta) + \varepsilon_{randomness}(\theta), \quad \text{s.t.} \tag{5a}$$

$$g_i(\theta) \leqslant 0 \quad i = 1, \ldots, n. \tag{6a}$$

where Θ is the design space, θ is a vector of the design space, $f(\theta)$ is the value of objective function at θ, $\varepsilon_{randomness}(\theta)$ is the inherent randomness in simulation outputs, $g_i(\theta)$ is the ith constraint of the problem and n is the total number of constraints.

As it was previously mentioned, the main duty of a metamodel is to construct a good approximation for the original function using a series of given inputs and the corresponding outputs. These inputs and outputs are usually obtained through Design of Experiment (DOE) methods. In fact, metamodels, which can be taken as abstraction models of a simulator, generally aim to reduce the model complexity, cost and time, yet maintain validity. If the function expressing the true nature of the computer simulation result is $y = f(x) + \varepsilon_{randomness}$, the metamodel is $y = \hat{f}(x) + \varepsilon_{approximation}$ in which $\varepsilon_{approximation}$ is the

Table 2
Validation of simulation model by comparison between actual and simulated data.

Severity index	Actual mean (min)	Simulated	
		Mean (min)	Confidence interval ($\alpha = 0.05$)
ESI 1	190	185.4	[157.4, 213.4]
ESI 2	203	211.8	[191.4, 232.2]
ESI 3, 4	210	219.6	[205.4, 233.8]
ESI 5	65	63	[57.6, 68.4]

Table 3
Predetermined parameters of mathematical model.

Decision variables	l_i	u_i	c_i (cost unit)
Triage nurses (x_1)	1	2	0.6
Receptionist (x_2)	1	3	0.5
Nurses (x_3)	1	6	0.8
Heart residents (x_4)	1	3	0.7
Beds (x_5)	5	10	2

fitting error. Therefore, by replacing the simulation model with a suitable metamodel, the respective optimization problem can be expressed as:

$$\min_{\theta \in \Theta} \hat{f}(\theta) + \varepsilon_{approximation}, \tag{5b}$$

$$g_i(\theta) \leqslant 0 \quad i = 1, \ldots, n. \tag{6b}$$

A schematic view of finding a suitable metamodel is illustrated in Fig. 3. We used this scheme to present the simulation-based optimization model which was fully presented in the next sections. As it is abovementioned, the objective of simulation is to minimize the total average waiting times of patients. Furthermore, we only considered some important resources in the ED such as triage nurses, receptionists, nurses, residents and beds which can directly and significantly affect the waiting time. However, the effect of other resources such as lab technicians and specialists is neglected for the following reasons:

- These resources are shared between different hospital's departments in addition to ED; therefore, ED's managers might not be able to directly control them.
- These resources do not directly affect or have a minor effect on the waiting times and therefore, not considering these resources for developing the simulation-based optimization model might significantly reduce the size of the design space. Furthermore, it could alleviate the process of fitting suitable metamodels by which more accurate and timely results could be obtained [27].

There are different methodologies in terms of mathematical formulation and the way of finding their parameters using the respective design points, however, three of these methods which have been widely used in the recent research studies are briefly reviewed and their strengths and weaknesses are briefly explained in the following sections. It is worth mentioning that reaching a proper metamodel could improve the accuracy and superiority of optimization results which can considerably influence the ED's performance.

3.2. Candidate metamodels

3.2.1. Response surface methods

Response Surface Models (RSM) were initially developed over fifty years ago to determine the optimal operating conditions in chemical processes [28]. All relevant studies usually prefer to use the lower order polynomials (e.g., the second order one) in which the pertinent coefficients are estimated by minimizing the total residual errors between the fitted values and the target values. Moreover, to assess the performance of the fitted metamodel, some indexes such as mean square error (MSE) are used. The mathematical formulation of the second order response surface is described as follows:

$$y = \beta_0 + \sum_{i=1}^{k} \beta_i x_i + \sum_{i=1}^{k} \sum_{j=1}^{k} \beta_{ij} x_i x_j + \varepsilon. \tag{4}$$

where k is the number of variables, β is regression coefficients and ε is error. The advantages of using RSM metamodels are the availability of known techniques for experimental designs, easy estimation of unknown parameters, and simple to interpret and assess. It is quite suitable and effective in a case study with a limited number of design variables (usually less than

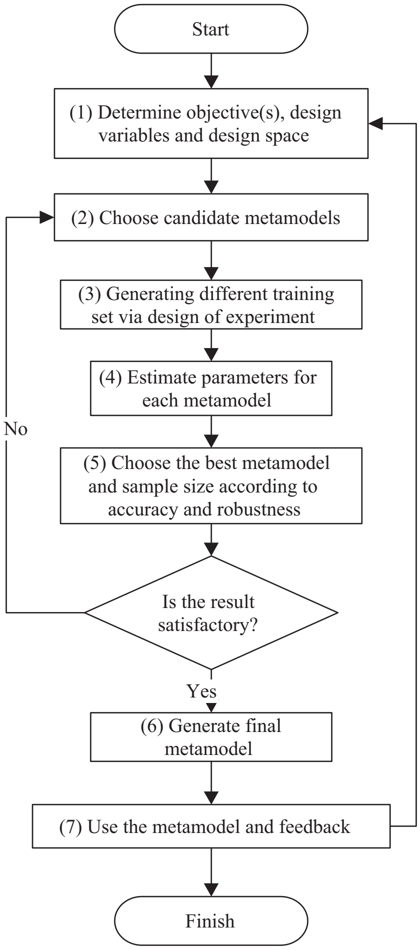

Fig. 3. Metamodeling process.

10) and with not-highly-nonlinear response [29]. Here we used Design Expert[1] software to find the values of the respective parameters in RSM.

3.2.2. Radial basis functions

There are multiple radially symmetric functions in Radial Basis Functions (RBF) which are centered at different points of design space. As the number of variables and the amount of nonlinearity in the responses is increased, the RSM becomes less applicable and RBF metamodels would be therefore a proper alternative technique [30]. To have a suitable RBF which appropriately represents the nonlinear response, the basic functions should be cautiously chosen. Although there are various basic functions and each of them has their own characteristics, Gaussian functions are widely used in the recent studies [31]. Furthermore, due to the diminishing influence of the radial points on the objective function by getting farther from the reference point, a Gaussian RBF would be a good choice for our application. Given k centers $c_1, c_2, \ldots, c_k \in \mathbb{R}^d$, the mathematical form of the radial basis function interpolation is described as follows:

$$y = \sum_{i=1}^{k} w_i \varphi(||x - c_i||) + \varepsilon. \tag{5}$$

where $||\cdot||$ is the Euclidean norm, φ is the Gaussian function, and $w_i \in \mathbb{R}$ for $i = 1, \ldots, k$ are coefficients. The RBF solver can be written in any programming language. We use MATLAB[2] software to do programming and to find the best values of the parameters.

[1] www.statease.com.
[2] www.mathworks.com.

3.2.3. Neural networks

Artificial Neural Networks (ANNs) are another popular type of metamodel which should be trained using a set of training data provided through simulation. ANNs have evoked great interest to be used as a metamodel because they are really powerful to learn the complex nonlinear functions in an efficient way [32–34]. Typical feedforward networks with one hidden layer and sufficient neurons in other layers can be used for any kind of input–output mapping problems. These networks usually consist of multiple layers in which the first layer has a connection from the network input. Each subsequent layer has a connection from the previous layer, and the final layer produces the network's output. A mathematical form of a network with three layers (i.e., one for input, another as a hidden layer, and the final layer for output) can be written as:

$$y = \sum_{j=1}^{l} w_j f\left(\sum_{i=1}^{k} v_{ij} f(x_i) + \alpha_j\right) + \beta + \varepsilon. \tag{6}$$

k is the number of variables, f is the transfer function defined by the user, v_{ij} is the weight of the connection between the ith input neuron and jth hidden neuron, α_j is the bias in the jth hidden neuron, w_j is the weight of connection between the jth hidden neuron and the output neuron, l is the total number of hidden neurons, β is the bias of the output neuron, and ε is the error. Three predetermined parameters of ANNs are: (1) the number of hidden layers (2) the number of neurons in each layer and (3) transfer functions. The Matlab Neural Network Fitting Tool was used for training the ANNs. Selection of parameters is usually based on trial-and-error and some simple rules of thumb.

3.3. Design of experiment

After choosing candidate metamodels, the training data should be prepared through one of the sampling methods (Design of Experiments) in the domain of decision variables such that these samples could cover all important parts of space. Sampling is a critical part of fitting a suitable metamodel as it must appropriately capture the system behavior against changes in the decision variables. The sampling plan (i.e., choosing a suitable design of experiment method and the size of sample) depends on the function under study, metamodel used and problem limitations. There are some research studies proposing a suitable design of the experiment method for each specific metamodel [35,36]; however, the size of samples are empirically determined in most cases.

The standard experimental design methods may be appropriate when the function is relatively smooth and the system involves less factors [37]. However, other complex methods such as Latin Hypercube Sampling (LHS), which can properly fill the respective space, are really recommended for those cases which have many factors and their underlying functions are unknown [38,39]. The criterion to find the points in LHS is to maximize the minimum distance between the design experiments with consideration of a constraint. This constraint maintains the spacing between factor levels. Moreover, using LHS, practitioners have many options to choose as design points and can more efficiently capture the non-linearity of the underlying system.

For our problem, three data sets including 20, 40, and 70 points are generated with the *lhsdesign(n,p)* function in Matlab that generates n design points for p variables. Furthermore, the corresponding objective function (the total average waiting time of patients) for each design point is obtained via simulation.

3.4. Parameters estimation

Given training data provided in the previous step, the coefficients of selected metamodels should be appropriately estimated. Table 4 illustrates the unknown parameters for each metamodel, their respective ranges and software used. Most of these parameters are fixed according to recommendations in the literature; however, the remaining parameters are determined in a trial-and-error approach.

3.5. Metamodel selection

To choose the appropriate metamodel and to find the suitable size of samples, two indicators including accuracy and robustness should be initially defined. The accuracy criterion, which is usually measured by the Root Mean Square Error $\left(rmse = \sqrt{\frac{1}{k}\sum_{i=1}^{k}(y_i - \hat{y}_i)^2}\right)$, indicates the average deviation of the metamodel outputs ($\hat{y}_i$) from the simulation output (y_i). To have a value between zero and one, we use $\left(\frac{1}{1+rmse}\right)$ instead. The robustness criterion is also measured using the standard deviation of one metamodel error value across different problems.

In the next step, we perform cross validation techniques to find the total error of all nominated metamodels and to choose the best of them using the above-calculated indicators. Here, we use the k-fold cross validation technique in which k is set to ten. It is worth mentioning that 10-fold cross validation is widely used, however, k is generally an unfixed parameter that should be predefined. In this type of cross validation, the training data set are split into ten subsets and the metamodel

Table 4
Summary of parameter estimation step for each metamodel.

Metamodel	Parameters	Range of parameters
RSM	1 – regression coefficients	–
RBF	1 – basis function	Gaussian
	2 – scaling factor (σ)	[0, 20]
	3 – coefficients	–
ANN	1 – number of hidden layers	1 or 2
	2 – number of neurons	4, 5, . . . , 8
	3 – transfer functions	Sigmoid – linear
	4 – training algorithm	Levenberg–Marquardt
	5 – coefficients	–

should be individually trained using each of these subsets and tested using nine of the remaining subsets (called validation subset). In other words, the metamodel is trained and tested ten times and total errors are calculated for each metamodel. For more explanation, one can refer to Meckesheimer et al. [40].

It is noteworthy to say that before constructing the metamodels, all the respective decision variables and the objective function should be scaled to state in the interval [0, 1]. This scaling makes all variables in the same order of magnitude; therefore, it might cause a significant improvement in the process of function (i.e., objective) approximation.

Values of "*rmse*" from 10-fold cross validation for the three selected metamodels with different sample sizes are illustrated in Fig. 4 and the detailed information is also summarized in Table 5. As it is obvious in this figure and table, ANN metamodels trained with 70 sample points have better performance in terms of both accuracy and robustness compared to other metamodels. However, as there are minor differences between the results of ANNs trained with 40 and 70 data points, we prefer to use ANN with less samples to reduce the number of evaluations which leads to reduction in running time.

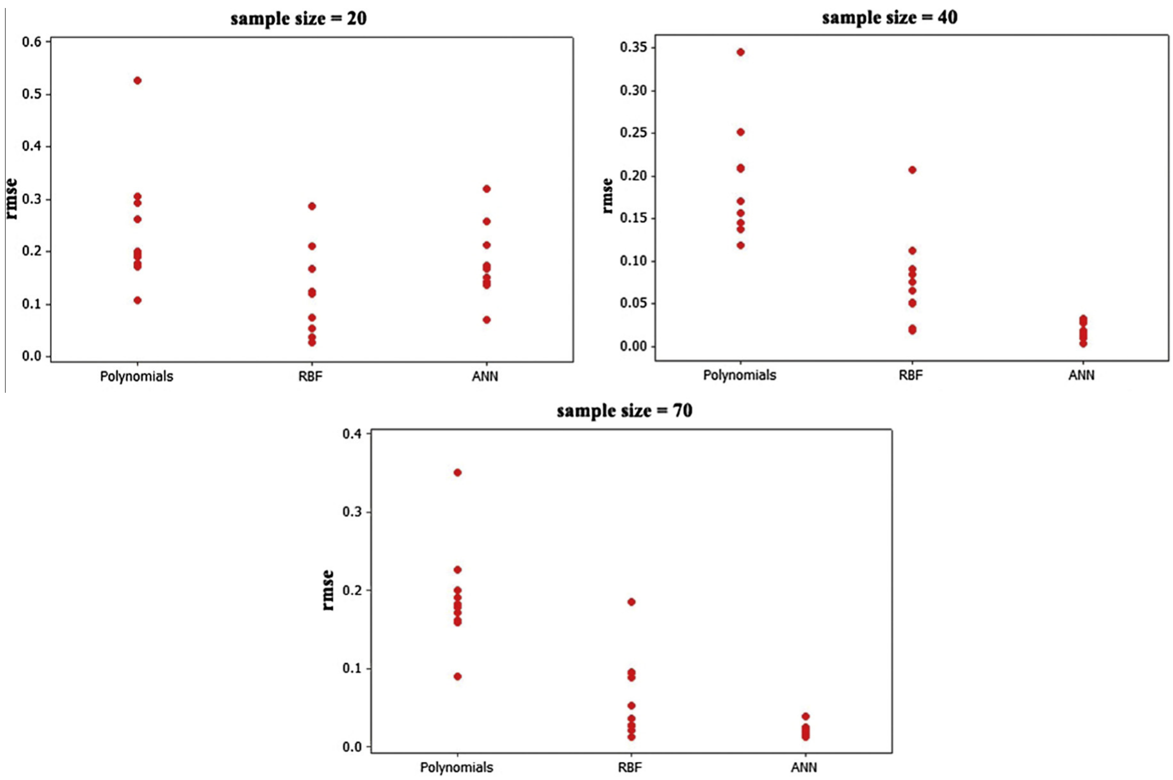

Fig. 4. Values of "*rmse*" from 10-fold cross validation for three metamodels with three sample sizes.

Table 5
Summary of 10-fold cross validation results for three metamodels in terms of accuracy and robustness.

Method	Error metrics for different LHS					
	3–20%		5–40%		10–70%	
	Accuracy	Robustness	Accuracy	Robustness	Accuracy	Robustness
Polynomials	0.695	0.116	0.832	0.071	0.861	0.031
Radial basis function	0.863	0.082	0.924	0.054	0.939	0.010
Neural network	0.786	0.069	0.977	0.009	0.980	0.0004

Here, the ANN's structure (i.e., the number of hidden layers and the number of neurons) is chosen on a trial-and-error basis. Therefore, we trained a network with one hidden layer and 5 neurons. The results which are obtained from the training process are illustrated in three graphs in Fig. 5 in which the first graph from the left demonstrates the "*mse*" error for each training, testing, and validation data set. Based on the correlation coefficient ($R = 0.99862$), target values and the output values of the ANN are very close for data consisting of all three data sets. The 20 bins error histogram also has a normal shape which shows that the ANN has balanced overestimations and underestimations of target.

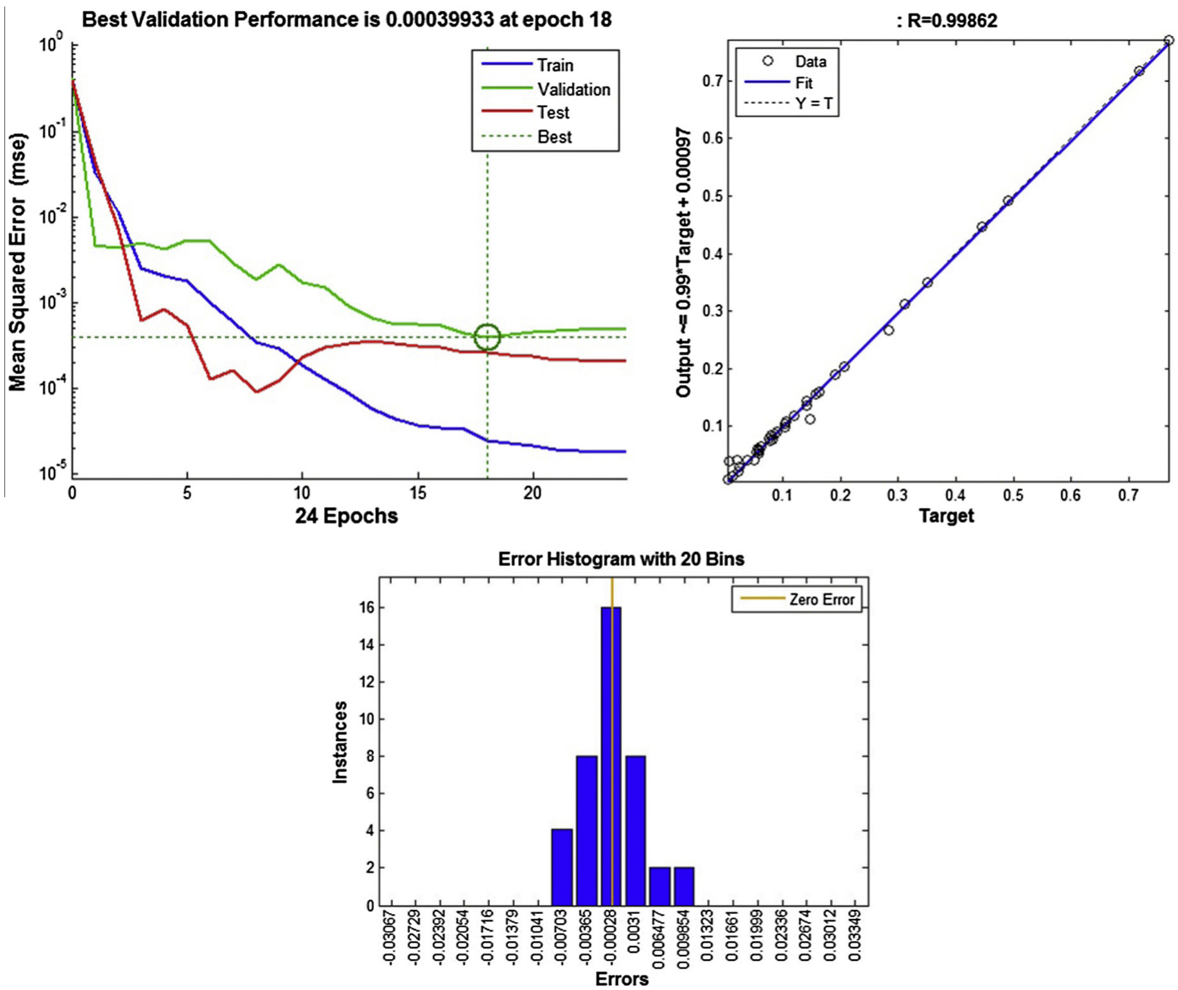

Fig. 5. Characteristics of the final ANN in training step.

4. Results and discussion

4.1. Proposed decision support system

The overall framework to use metamodel-based optimization for resource planning is shown in Fig. 6. This figure illustrates how we can incorporate the simulation metamodel into a DSS. The system user initially determines when the optimization process should be implemented. In the next step, the system status should be reported in an appropriate way. The user must analyze the system in addition to constructing the simulation model with a suitable metamodel. Based on the user objectives and the information extracted from the system, the user should update the simulation model. If there is a need for finding optimum configuration of the resources (e.g. changes in service rate, changes in the process, etc.) the optimization is then carried out with combination of metamodel and simulation, and finally, the optimal solutions are returned back to the simulation model for verification. This can be an iterative process to slowly building up the accuracy of the metamodel. After making the proper decision, it is time to take action. The proposed decision support system can be used for creating guidelines in strategic, tactical, and operational management practices. Some issues related to the operational decision-making is introduced in Section 5 because this is more new and challenging topic.

4.2. Optimization

By replacing the objective function with the ANN metamodel in our mathematical model described in Section 3.5, a new approximate model is used to determine the best design of resources by considering a total budget of 18.7. This design is a very crucial decision for hospital managers because the waiting time for patients with heart problems should be as short as possible. As it is illustrated in Table 6, our new methodology obtains a different resource design compared to what is currently performed for this department. Our approximate model suggests that one nurse be cut back and one resident be added to the department. These changes cause about 48% improvement in the total average waiting times of patients (i.e., reducing the time from 44 to 23 min)

Table 7 also demonstrates changes on the queues in the different sections of the department. As it is illustrated in this table, more improvements occurred on the resident and bed queues. Adding one resident changed the waiting time of residents from 18.582 to 1.336 min. Because of overall improvement in patients flow and patient throughput in the ED, the waiting times for beds is decreased from 15.66 to 9.888 min.

Furthermore, the set of graphs in Fig. 7 demonstrates the amount of sensitivity of the approximate model to the change in each decision variable (i.e., the number of resources in each section) in terms of total average waiting times of patients. To carry out this task, we change the value of a single decision variable around its optimum point (optimum level is illustrated by empty circle) while others are fixed at their optimal values. As it can be seen in these graphs, except for the triage nurse, adding other resources does not remarkably decrease the total average waiting time.

4.3. Metamodel verification

To verify the efficiency of the approximate model constructed based on metamodels, we compare the results with those obtained from OptQuest[3] in terms of accuracy and time needed for optimization. OptQuest is a commercial tool which is linked with simulation softwares like the arena and can perform simulation-based optimization for different applications. It combines multiple metaheuristics into a single heuristic methodology to search the respective design space in a systematic way [41]. Moreover, exhaustive search, as a test bed, is performed by which a suitable upper bound can be obtained for both metamodel and OptQuest methods. We perform $2 \times 3 \times 6 \times 3 \times 6 = 648$ experiments on the design space to carry out exhaustive enumeration. Table 8 reports the optimization results for the three methodologies above with different budget limitations.

As it is observed in this table, OptQuest has better total average waiting times compared to the metamodel-based method in terms of accuracy because OptQuest does not have approximation error. The promising point for a metamodel is the time needed for optimization. All the experiments are implemented on a system with Intel core i7-2630QM Processor 2 GHZ and 4 GB RAM. Each experiment takes about 45 s (for batch run mode of the arena). Therefore, it takes about 486 min if all possible solutions are experimented in the exhaustive enumeration. This might be very long especially where it is going to be used at a level of operational decision-making. However, using the metamodel-based methodology implemented with forty experiments, we can have a very close solution to an optimal one while it only takes about 30 min. It can be a promising outcome especially where this methodology is used in real-life health applications with a huge design space.

This is also worth mentioning that if the size of solution space and the number of evaluations increase, the time needed to find the optimal solution might dramatically grow in OptQuest. Table 9 illustrates the number of simulation needed to find the optimum design and the corresponding time needed for various design spaces where OptQuest is applied. To provide a better comparison, we do not change the initial solution at each run of the OptQuest algorithm.

[3] www.opttek.com.

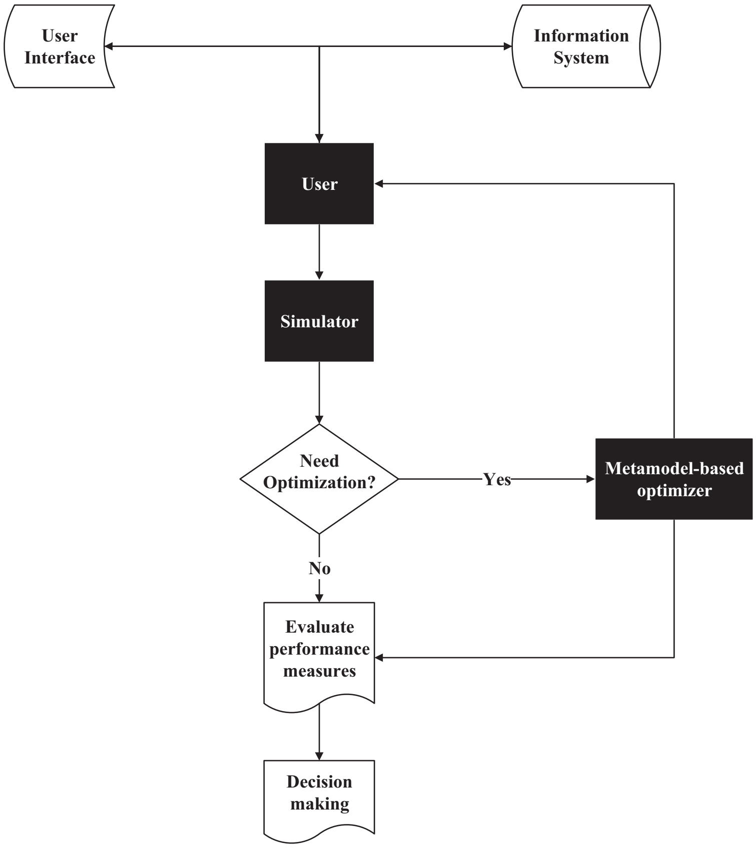

Fig. 6. Metamodel-based decision support system.

Table 6
Comparison between current design and optimal design.

Design	x_1	x_2	x_3	x_4	x_5	Budget	Total average waiting times (min)
Current design	1	2	3	1	7	18.7	44
Optimum design	1	2	2	2	7	18.6	23

Table 7
The impact of new design on waiting times of the queues.

Queues	Total average waiting time (min)	
	Current design	Optimum design
Triage nurse	7.134	7.044
Receptionist	2.001	2.001
Nurse	1.056	2.272
Bed	15.66	9.888
Resident	18.582	1.336

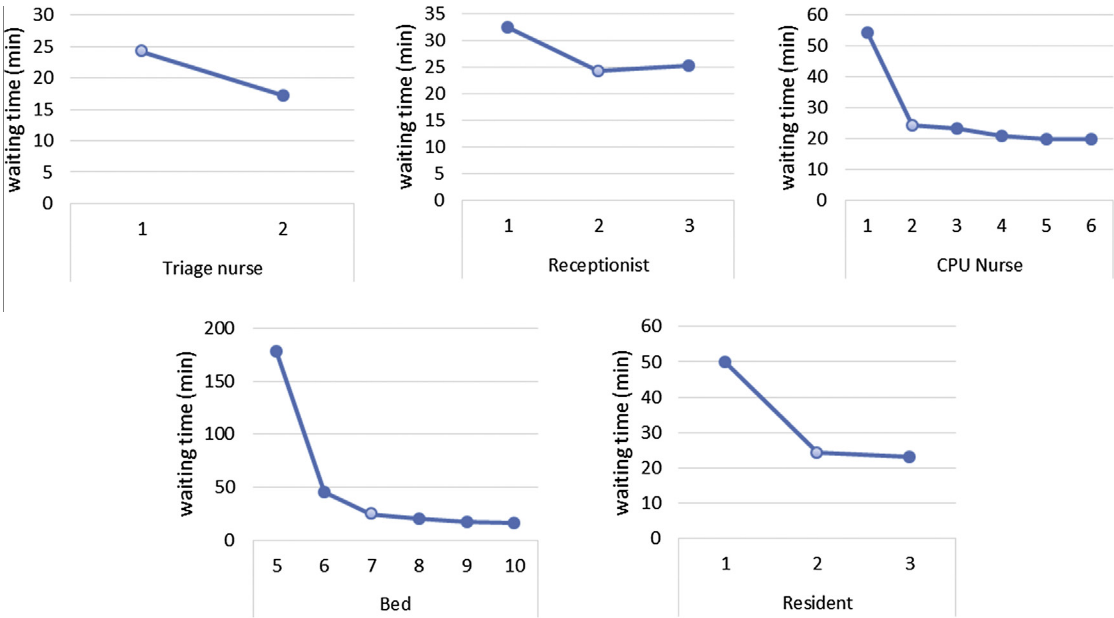

Fig. 7. Sensitivity of total average waiting times of patients to variations in resource levels.

Table 8
Verification of the metamodel against OptQuest and exhaustive search methodologies.

Budget values (B)	Methods																				
	Metamodels							OptQuest							Exhaustive Search						
	x_1^*	x_2^*	x_3^*	x_4^*	x_5^*	*Cost*	f^*	x_1^*	x_2^*	x_3^*	x_4^*	x_5^*	*Cost*	f^*	x_1^*	x_2^*	x_3^*	x_4^*	x_5^*	*Cost*	f^*
20	2	2	2	3	7	19.9	15	2	2	2	3	7	19.9	15	2	2	2	3	7	19.9	15
21	2	2	3	3	7	20.5	13	2	2	3	3	7	20.7	13	2	2	3	3	7	20.7	13
22	2	3	2	2	8	21.7	11	2	2	3	2	8	22	9	2	2	3	2	8	22	9
23	2	3	3	2	8	22.5	9	2	2	3	3	8	22.7	7	2	2	3	3	8	22.7	7
24	2	3	2	2	9	23.7	7	2	2	3	2	9	24	6	2	2	3	2	9	24	6
25	2	2	3	3	9	24.7	5	2	2	3	3	9	24.7	5	2	2	3	3	9	24.7	5
26	2	3	3	4	9	25.9	4	2	3	3	4	9	25.9	4	2	3	4	3	9	26	4
27	2	2	3	3	10	26.7	3	2	2	3	3	10	26.7	3	2	3	5	3	9	26.8	2
29.6	2	3	5	3	10	28.8	2	2	3	5	3	10	28.8	2	2	3	6	3	10	29.6	<1

Table 9
Number of simulation needed to find the optimum point and the corresponding time.

Budget value	20	21	22	23	24	25	26	27	29.6
Design space	268	321	376	429	484	536	580	613	648
Number of simulated points in OptQuest	146	159	160	179	185	185	195	204	215
Simulation time (min)	109.5	119	120	134	139	139	146	153	161

5. Challenging issues for operational resource planning

Before we can create a user-friendly decision support system even for personnel with minimal operations research experience and before finding the optimal level of resources which can be really helpful for ED managers in maintaining the quality of services at some constant level, we should answer the following questions:

- By what mechanism can we monitor the ongoing operations in the ED?
- Which parameters (e.g., input rates, service rates, queue length, etc.) should be monitored?
- How can we use these data to predict the operational state of the ED for several hours later into the future?
- How can we carry out prediction analysis within a short time?
- What is the definition of optimal condition?
- How can we actually change the number of resources at an operational level?

Having enough information about the current state of the ED is crucial for creating a real-time DSS. This information might include: (1) The physical location of people and equipment (2) The number of patients in the different queues (3) The utilization and status of resources (4) The bottlenecks that may affect throughput of the patients.

With regard to existing infrastructures in Modarres emergency center, the above information cannot be easily achieved as data collection including database queries from Hospital Information System (HIS) is currently performed on a traditional basis. Therefore, the needed data might not be sufficiently accurate and mostly not be available in detail. In other words, we only access the respective data pertinent to patients who enter the reception stage which does not have useful information about the current state of the ED. Thus we need to focus on the new technologies like Radio Frequency Identification (RFID) which is successfully used in some EDs [42,43]. RFID is the wireless use of radio-frequency electromagnetic fields to transfer data for the purposes of automatically identifying and tracking tags attached to objects. Using these technologies we can monitor the current state of the system and extract the required information [44].

Since congestion and thus an increase in waiting times of patients occurs with some delay after changing the system parameters [45], with rapid estimation of system parameters and predicting the state of the system a few times in the future, we can make proper decisions on the configuration of resources and hence carry out proper action. In other words, to change the number of resources (staffs and beds) at an operational level and balance the patient flow in the system we should transfer resources from idle stations to more crowded stations and use on-call staffs.

The proposed conceptual framework for determining the optimum number of resources in order to maintain optimal conditions is as follows:

(1) System status should be observed by the user and the system controller. In addition to monitoring, construction and completion of models and analysis of the system is the user's task.
(2) Based on information extracted from previous steps, the user should predict the situation of ED for the next few hours.
(3) If required, the optimal number of resources should be calculated and the status of the current state and the optimized one are compared with each other via simulation. (Based on the size of solution space and the time required for the simulation, the user can carry out the optimization with or without a metamodel).
(4) After making the proper decision, it is time to take action.

6. Conclusion and future research

In this paper, an attempt is made to extend the application domain of metamodel-based optimization to healthcare managerial problems. As it could be observed from the literature, the combination of both simulation and metamodels is a state-of-the-art subject that can be used in different applications such as healthcare systems. The respective model stated in the core of a flexible decision support system which can produce a fast and accurate solution and even provide management guidelines to enhance the performance of emergency departments. It could remarkably assist managers to make operational, tactical and strategic decisions on their systems.

We demonstrated that how metamodels can be applied to find a near-optimal resources allocation which can dramatically reduce the patient congestion in the emergency departments. In other words, we developed a decision support including a metamodel-based optimization to obtain a configuration of resources which could remarkably reduce the total average waiting times of patients. We implemented our proposed methodology in Modarres Emergency Center located in the province of Tehran, Iran.

To construct an appropriate metamodel which can discover the highly nonlinear curvature of the response of the healthcare system, we developed a computer simulation model by which the total waiting time of patients can be estimated for each design point (i.e., the number of receptionists, nurses, residents, and beds). Using these sample points and a well-known cross validation technique, we chose an appropriate metamodel among three popular ones (i.e., Artificial Neural Network, Radial Base Function, Response Surface Model). It was demonstrated that ANN trained with 40 sampling points in Modarres Hospital had better performance compared to the two other metamodels in terms of both predefined criteria: accuracy and robustness. By extracting an appropriate ANN, the optimization process is carried out to find the near-optimal resources considering the budget and other respective constraints.

The promising point is that by implementing the optimum allocation obtained from proposed DSS, in the emergency department of Modarres Hospital, we reached a 48% decrease in total average waiting times of patients. To illustrate the validation of our metamodel-based optimization, we compared our results with those obtained from OptQuest in terms of both accuracy and time needed to finish the optimization process. The comparison verified that the efficiency of our proposed model was sufficiently acceptable. However, the proposed methodology can be efficiently applied for other real-life healthcare problems with the capability of creating a high qualified solution. We really believe that this research could be a good start for other healthcare applications which can be similarly modeled and solved.

It should be also noted that the metamodel-based optimization is used for obtaining the tactical level of decision-making while it could be used for other level such as operational level. Other metamodels (e.g., Genetic Programming) and different designs of experimental methods could be used to improve the results. Furthermore, we constructed and implemented the respective model for emergency department without consideration of its relations to other departments of the Modarres Hospital. Extending this methodology for modeling the other sections of the hospital or the whole hospital could be also other areas of research which are all left for future studies.

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Appendix A

Training data set.

Scenario	Decision variables					Total average waiting times of patients (normalized)
	Triage nurses	Receptionists	Nurses	Heart residents	Beds	
1	2	2	5	6	3	0.10214
2	1	1	1	9	3	0.179438
3	1	3	5	7	2	0.071739
4	2	3	5	10	1	0.095061
5	2	2	5	7	1	0.131095
6	1	1	6	7	1	0.179208
7	2	3	2	8	3	0.023589
8	1	1	1	8	2	0.187899
9	2	2	6	9	3	0.004807
10	2	3	4	10	1	0.100001
11	1	2	1	6	3	0.264122
12	1	3	1	5	1	1
13	1	1	1	7	3	0.206909
14	2	2	2	8	2	0.027612
15	1	3	6	7	3	0.064605
16	2	1	2	10	1	0.152007
17	1	3	4	5	3	0.453303
18	2	1	6	10	2	0.061113
19	1	2	2	9	1	0.132574
20	1	3	5	8	3	0.046001
21	2	3	3	6	3	0.097013
22	2	2	3	5	2	0.399806
23	2	2	5	8	2	0.021764
24	1	1	3	5	1	0.932607
25	2	1	1	9	1	0.242316
26	1	1	6	6	1	0.365503
27	2	3	4	6	1	0.199158
28	2	1	4	10	2	0.060461
29	2	2	3	6	3	0.106434
30	1	1	2	9	2	0.07201
31	1	2	6	5	2	0.575472
32	1	2	2	7	2	0.075958
33	2	3	4	10	3	0
34	1	2	4	7	3	0.069589
35	2	3	6	6	1	0.200999
36	2	3	2	8	1	0.111898
37	1	1	2	9	3	0.068292
38	2	2	3	9	2	0.012084
39	1	3	4	10	1	0.129855
40	1	1	3	5	2	0.635259

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